IMO(18^{th}) – 1976

First Day

1. In a convex quadrangle with area $32cm^2$, the sum of the lengths of two nonadjacent edges and of the length of one diagonal is equal to 16 cm. What is the length of the other diagonal?
2. Let $P_1(x) = x^2 - 2$, $P_j(x) = P_1(P_{j-1}(x))$, $j = 2,3,\ldots.$ Show that for arbitrary n, the roots of the equation $P_n(x) = x$ are real and different from one another.
3. A rectangular box can be filled completely with unit cubes. If one places the maximal number of cubes with volume 2 in the box such that their edges are parallel to the edges of the box, one can fill exactly 40% of the box. Determine all possible (interior) sizes of the box.

Second Day

4. Find the largest number obtainable as the product of positive integers whose sum is 1976.
5. Let a set of p equations be given,

   $a_{11}x_1 + a_{12}x_2 + \ldots + a_{1q}x_q = 0,$

   $a_{21}x_1 + a_{22}x_2 + \ldots + a_{2q}x_q = 0,$

   ……

   $a_{p1}x_1 + a_{p2}x_2 + \ldots + a_{pq}x_q = 0,$

   With coefficients $a_{ij} = -1,0,1$ for all $i = 1,2,\ldots,p$ and $j = 1,2,\ldots,q$. Prove that if $q = 2p$, there exists a solution $x_1, x_2, \ldots, x_q$ of this system such that all $x_j\ (j = 1,2,\ldots,q)$ are integers satisfying $|x_j| \geq q$ and $x_j \neq 0$ for at least one value of j.
6. For all positive integer n, $u_{n+1} = u_n(u_{n-1}^{\ 2} - 2) - u_1$, $u_0 = 2$ and $u_1 = 2\frac{1}{2}$. Prove that $3\log_2[u_n] = 2^n - (-1)^n$, where $[x]$ is the integral part of *x*.

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